

Chest X-Ray Pneumonia Screening — CNN Comparison Demo

Research Methodology, Model Evaluation & Explainability Case Study

RESEARCH DEMONSTRATION ONLY — NOT A MEDICAL DEVICE.
This tool has not been validated for clinical use or cleared by any regulatory authority (FDA/CE). It is trained on a limited pediatric dataset (n=5,856) from a single source institution. It should NOT be used to make diagnostic or clinical treatment decisions.

Project Overview & Methodology

This project evaluates convolutional neural network (CNN) architectures trained via transfer learning to classify anterior-posterior chest radiograph scans into two categories: NORMAL or PNEUMONIA. Training was conducted on the Kermany et al. Chest X-Ray Images dataset, consisting of 5,856 pediatric scans preprocessed to match standard inputs. Images are converted to grayscale, replicated to 3 identical channels, resized to 224x224, and standardized using ImageNet mean and standard deviation parameters.

Model Performance Metrics

Architecture	Accuracy	F1-Score	AUC-ROC	Explainability Diagnostic Focus (Grad-CAM)
ResNet18 (Deployed)	87.8%	0.911	0.978	Non-diagnostic artifacts (Shoulder / Collarbone / Labels)
MobileNetV2	85.6%	0.896	N/A	Anatomical lung regions (Pulmonary fields)
EfficientNet-B0	84.8%	0.891	N/A	Diffuse lung markings and surrounding tissues

Core Finding: The Interpretability Trade-off

Grad-CAM (Gradient-weighted Class Activation Mapping) explainability analysis revealed an critical safety finding: **the model with the highest accuracy (ResNet18, 87.8%) is not the most clinically safe or explainable.** In contrast to MobileNetV2 which targets active pulmonary parenchyma directly, ResNet18's attention heatmaps consistently focus on non-diagnostic skeletal structures (the shoulder joint, collarbone, and positioning artifacts). This indicates ResNet18 learned shortcut features linked to patient posture and cropping templates common in pneumonia X-rays, rather than pathologically relevant lung consolidation. MobileNetV2 (85.6% accuracy), although scoring lower, displays a highly localized diagnostic focus on lung fields, representing a safer option for clinical integration.

Technical & Clinical Limitations

- Single-Institution Pediatric Cohort:** The Kermany et al. dataset originates entirely from a single medical center (Guangzhou Women and Children's Medical Center) and consists exclusively of pediatric patients. Performance is not generalized to adult cohorts or diverse imaging hardware.
- No External Dataset Validation:** Models have not been validated on external datasets. Out-of-distribution performance (from different clinics, x-ray models, or patient postures) is unquantified.
- Qualitative Explainability:** Grad-CAM heatmaps represent a qualitative, investigator-led check. No systematic quantitative clinical trial was performed to evaluate radiologist alignment.
- No Regulatory Approval:** This demonstration is strictly for educational and student portfolio presentation. It has not been approved by the FDA, CE, or any other body for diagnostic use.

PneumoScan Research Project Summary Sheet. Prepared for presentation and discussion purposes. Code repository: github.com/Prathamesh250205/PneumoScan-CNN. Live demo: pneumoscanner.onrender.com.